

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built operational SaaS automation for multi-step filings where each browser/API action recorded recoverable state, validation errors, and operator checkpoints.
- Reduced manual filing time by **85%** by hardening flaky public-site workflows with retries, session recovery, and backend state validation.
- Used Docker, Cloud Run, and GitHub Actions to keep deployments repeatable and reduce release friction by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Improved reliability and performance inside a large open-source Python application with image-processing, model-output, and visualization workflows.
- Reduced runtime by **87%** by splitting expensive processing into asynchronous and parallel paths.
- Debugged mature-codebase issues across UI, data transformation, generated artifacts, and PyTorch/ONNX-adjacent integration points.

Open Source

Contributions 

- Contributed maintainer-reviewed patches across **Tiled**, **Kuadrant/Kyverno**, VideoLAN, CRIU, and IOOS, spanning UI/compatibility fixes, backend integration behavior, parser robustness, and Zip Slip path validation.

Projects

Hyoka | *Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL, Next.js, TypeScript, Docker*

- Built a **reliability control plane for AI agents** with trace/event ingestion, authenticated APIs, suites, candidates, validation runs, gate decisions, promotions, audit logs, and a Next.js operations dashboard.
- Implemented SQLAlchemy/Alembic metadata models, project-scoped API keys, content-addressed artifacts, signed manifests, worker-owned execution leases, and Docker Compose deployment with Postgres-backed services.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

GitHub 

- Built browser-automation-backed product workflows with REST APIs, RabbitMQ workers, Redis autosave, PostgreSQL persistence, durable job status, and recoverable external publishing tasks.
- Added retries, DLQs, circuit-breaker isolation, health checks, and Prometheus metrics so long-running API/browser workflows stayed observable and fault-tolerant.

Swish Support System | *JavaScript, React, Python, PostgreSQL/pgvector, Redis, Langfuse*

GitHub 

- Built a hybrid AI + rules support system where LLM assessments classify messy complaint semantics while deterministic policy code controls refunds, replacements, coupons, and escalations.
- Grounded responses in order, item, evidence, customer, and policy context with PostgreSQL/pgvector, Redis state, Langfuse traces, confidence thresholds, and regression checks.

Technical Skills

Languages: JavaScript, TypeScript, Python, Go, SQL

Frameworks: React, Next.js, Node.js, Express, FastAPI, Echo

Tools: PostgreSQL, Redis, Docker, GitHub Actions, GCP, Prometheus, Langfuse, Postman, Firebase